



**MANIPAL UNIVERSITY
JAIPUR**

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SEMESTER I

AI6101: APPLIED STATISTICS AND PROBABILITY

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Basics of Statistics

Statistics is the science of collecting, organizing, analysing, and interpreting data to uncover patterns and make informed decisions. It helps in understanding variability, drawing conclusions, and making predictions based on data from samples or populations.

Basics of Statistics

- **Statistics** is the science of collecting, organizing, analyzing, and interpreting data to make informed decisions.
- **Types of Statistics:** Descriptive (summarizes data) and Inferential (draws conclusions from data).
- **Data Types:** Qualitative (categorical) and Quantitative (numerical).
- **Measures of Central Tendency:** Mean (average), Median (middle value), and Mode (most frequent value).

Basics of Statistics

- **Measures of Dispersion:** Range, Variance, and Standard Deviation show data spread.
- **Probability:** Measures the likelihood of an event occurring, forming the foundation of inferential statistics.
- **Normal Distribution:** A bell-shaped curve where most data falls near the mean.
- **Sampling:** Selecting a subset from a population to make estimates or test hypotheses.

The Role of Statistics in Engineering

- **Quality Control:** Ensures product consistency and reliability through statistical process control (SPC) and Six Sigma.
- **Design of Experiments (DOE):** Helps in optimizing product designs and processes by testing multiple variables efficiently.
- **Reliability Engineering:** Uses statistical models to predict product life and failure rates.
- **Data Analysis:** Interprets experimental and operational data to improve systems and processes.

The Role of Statistics in Engineering

- **Risk Assessment:** Quantifies uncertainty and helps in making risk-informed engineering decisions.
- **Simulation & Modeling:** Supports engineering designs with probabilistic models and simulations.
- **Decision Making:** Aids in choosing optimal solutions using statistical decision theory.
- **Process Improvement:** Identifies root causes of variation and supports continuous improvement initiatives.
- [role-of-statistics-in-engineering-a-review-IJERTV11IS100057.pdf](#)

Basic Principles

- **Data Collection:** Accurate and relevant data must be collected systematically.
- **Organization:** Data should be arranged in tables, charts, or graphs for clarity.
- **Summarization:** Use measures like mean, median, and mode to summarize data.
- **Variation Analysis:** Understand how data values vary using range, variance, and standard deviation.
- **Probability Concepts:** Apply probability to measure uncertainty and predict outcomes.

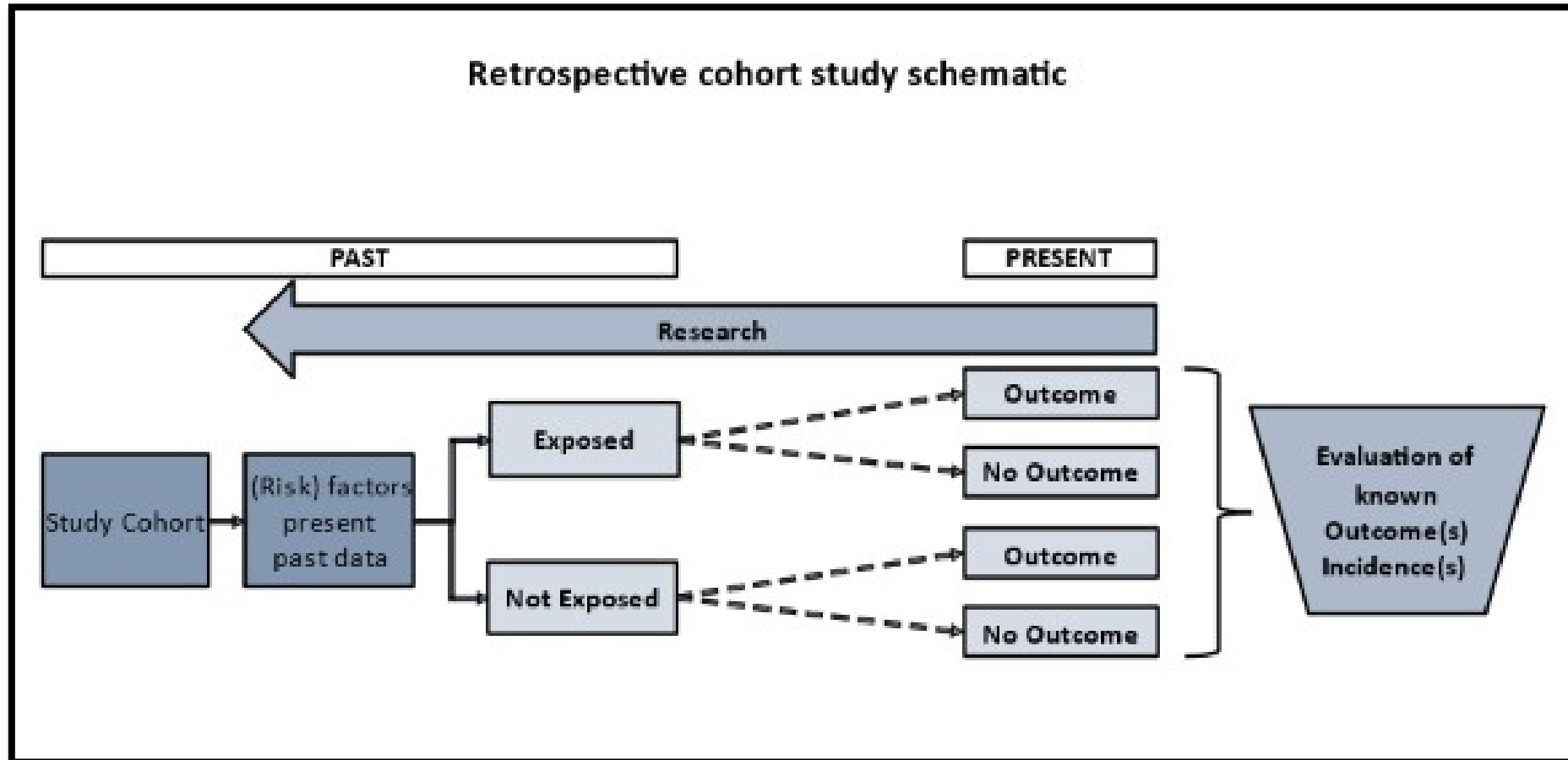
Basic Principles

- **Sampling:** Use representative samples to draw conclusions about a larger population.
- **Inference:** Make predictions or decisions based on sample data using inferential statistics.
- **Estimation:** Estimate population parameters (e.g., population mean) using sample statistics.
- **Hypothesis Testing:** Test assumptions or claims about a population using data.
- **Objectivity:** Maintain impartiality and avoid bias throughout the statistical process.

Retrospective Study

- A retrospective study analyzes past data or records.
- It studies causes or risk factors after the outcome has already happened.
- It is usually designed as a case-control study.
- It is quick and cost-effective because it uses existing data.
- It shows associations, not direct cause-and-effect.
- It may suffer from recall bias or selection bias.
- Ethical approval is usually easier to get.
- It is commonly used in healthcare, epidemiology, and public health

Retrospective Study



Observational study

- An observational study involves watching and recording data without interfering.
- Researchers do not manipulate variables or assign treatments.
- It is used to find natural associations between variables.
- There are two main types: retrospective (past data) and prospective (future tracking).

Observational study

- Common in fields like medicine, psychology, and social sciences.
- It can identify correlations but cannot prove causation.
- Usually easier and cheaper than experimental studies.
- Useful when experiments are unethical or impractical.

Designed Experiments

- In a designed experiment, researchers actively control and manipulate one or more variables.
- Participants are usually assigned to treatment and control groups.
- The goal is to determine cause-and-effect relationships.
- It includes elements like randomization, replication, and control.

Designed Experiments

- Designed experiments reduce the chance of bias and confounding variables.
- Common in scientific research, agriculture, medicine, and engineering.
- It is more accurate but often more time-consuming and expensive.
- Helps establish clear conclusions about the effect of interventions.

Observing Processes Over Time

- Observing processes over time means tracking changes in a system or activity continuously or at regular intervals.
- It helps identify trends, patterns, and fluctuations in behavior or performance.
- This method is often used in quality control, manufacturing, healthcare, and scientific research.
- Tools like control charts, run charts, and time series plots are commonly used.

• It allows early detection of problems or abnormalities in a process.

Observing Processes Over Time

- Long-term observation helps in making data-driven decisions.
- It is essential for process improvement and predictive analysis.
- The goal is to maintain consistency, efficiency, and quality over time.
- It involves collecting and analyzing data in a systematic way.
- Often combined with statistical methods for better

Mechanistic and Empirical Models

Mechanistic Models

- Mechanistic models are based on known scientific laws or theories.

- They explain how a system works using cause-and-effect relationships.

- These models use mathematical equations to represent physical, chemical, or biological processes.

- They are highly interpretable and useful for understanding system behavior.

Mechanistic and Empirical Models

Empirical Models

- Empirical models are built from observed data, not from physical laws.
- They focus on finding patterns or correlations in data.
- These models often use statistical or machine learning techniques.
- They may provide good predictions but offer less understanding of underlying mechanisms.
- **Example:** Linear regression predicting sales from past trends.

Probability and Probability Models

Probability

- Probability is a measure of how likely an event is to happen.
- It ranges from 0 (impossible) to 1 (certain).
- It helps in quantifying uncertainty in real-life situations.
- Example: The probability of getting heads in a coin toss is 0.5.
- Probability is the foundation of statistics, AI, and data science.

Probability and Probability Models

Probability Models

- A probability model is a mathematical representation of a random process.
- It includes a sample space, events, and probabilities assigned to those events.
- It helps predict how often outcomes occur over many trials.
- Examples: Binomial, Poisson, and Normal distributions.
- These models are used in simulations, decision-making, and risk analysis.

Thank You....